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A Appendix

Genuinely ambiguous annotation instances.

As mentioned in the main text, we analysed 100 examples with high disagreement among annotators and found that the majority of the case are truly ambiguous and cannot be assigned a definitive True/False label. Here we show two concrete examples. In Fig. 13, we can infer that the keyboard is horizontally lower than the cat. However, the keyboard is not right under the cat’s body (but a bit in front of the cat). The ambiguity results from the semantics of the word “below” which can mean both (1) horizontally lower (but not necessarily right under) and also (2) right under. In Fig. 14, if seat surface is viewed as the main body of the chair, we can infer that the sandwich is indeed above the chair. However, it is not above all parts of the chair (it is no higher than the backrest of the chair). This case depends on a person’s subjective understanding of what counts as “above the chair” and is thus also intrinsically ambiguous.



Figure 13: Caption: *The keyboard is below the cat.*



Figure 14: Caption: *The sandwich is above the chair.*

While we have excluded such cases in the standard VSR train/val/test split to ensure the dataset has human consensus, we also release the raw dataset without excluding these ambiguous cases for researchers who are interested in studying them further.

Relations included and excluded. Our used set of relations is adapted from Marchi Fagundes et al. (2021). In Table 6, we copy their original table for reference. We also list the exact modifications we made on top of the original table in Table 7. The major reasons for excluding relations are: (1) rarely used in describing spatial information of images (e.g., “north of”), (2) repeated with other relations (e.g., “front of” and “in front of”). In order to convert relations into templates, we also inserted some words to make them grammatical. We included several additional relations since we found them very frequent in MS COCO (e.g., “at the edge of” and “touching”).

Screenshots of the annotation interface. We use the open-sourced label studio (labelstudio.io) for managing our annotation tasks. Two screenshots are shown in Fig. 15 (caption generation) and Fig. 16 (validation).

Category	Spatial Relations
Adjacency	Adjacent, alongside, side, right side, left side, attached, back side, front of, ahead, against
Cardinal direction	North of, south side of, east-west, easterly, north-south, south of
Directional	Off, past, toward, down, deep down, up, away from, along, around, from, into, to, across, across from, through, down from
Orientation	Facing, orientation, parallel, perpendicular
Projective	On top of, beneath, in back of, beside, behind, left of, right of, under, in front of, below, above, over
Proximity	By, close to, near, far, far from, far away from
Topological	Congruent, connected, detached, has part, part of, contains, within, at, on, in, with, surround, among, consists of, out of, between, inside, outside
Unallocated	Beyond, next to, opposite, after, among, enclosed by

Table 6: The original table from [Marchi Fagundes et al. \(2021\)](#).

Category	Spatial Relations
Adjacency	Adjacent to , alongside, at the side of , at the right side of , at the left side of , attached to , at the back side of , front of , ahead of , against, at the edge of
Cardinal direction	North of , south side of , east-west , easterly , north-south , south of
Directional	Off, past, toward, down, deep down, up, away from, along, around, from, into, to, across, across from, through, down from
Orientation	Facing, facing away from , orientation , parallel to , perpendicular to
Projective	On top of, beneath, in back of , beside, behind, left of, right of, under, in front of, below, above, over, in the middle of
Proximity	By, close to, near, far , far from, far away from
Topological	Congruent , connected to , detached from , has as a part, part of, contains, within, at, on, in, with, surrounding , among, consists of, out of, between, inside, outside, touching
Unallocated	Beyond, next to, opposite to , after, among, enclosed by

Table 7: Our used table. **What we added.** **What's deleted.**